

Technical Document #485: Computer Vision - Comprehensive Overview

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Visual question answering (VQA) answers questions about images. It requires both visual understanding and language comprehension.

Image super-resolution reconstructs high-resolution images from low-resolution inputs. GAN-based methods produce photorealistic upscaled images.

Image classification assigns a label to an image from a fixed set of categories. CNNs like ResNet and EfficientNet achieve superhuman performance on benchmark datasets.

Face recognition identifies or verifies faces in images. DeepFace and FaceNet achieve near-human accuracy on face verification tasks.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Pose estimation detects human body keypoints in images.
- Image classification assigns a label to an image from a fixed set of categories.
- Optical character recognition (OCR) converts images of text into machine-readable text.